University of Heidelberg

Faculty of Mathematics and Computer Science
Institute of Computer Science
Research Group: Data Analysis and Modeling in Medicine

Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

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Abstract

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1.3. Research Questions

The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

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6.1. Conclusion

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Acknowledgment

References

[1] Aymeric Fleith, Doaa Ahmed, Daniel Cremers, and Niclas Zeller. Lifcal: Online light field camera calibration via bundle adjustment. https://arxiv.org/abs/2408.11682, 2024.